

**DESCRIPTION OF COURSEWORK**

Course Code	CST207
Course Name	Design and Analysis of Algorithms
Lecturer	Dr. Draib Najla'a Ateeq Mohammed Dr. Mohammed N. M. Ali Dr. Geetha Kanaparan
Academic Session	2026/04
Assessment Title	StudySmart AI - Project (35%)

A. Introduction/ Situation/ Background Information

In university life, students often have many tasks to complete before exams, such as revising lectures, completing tutorials, finishing assignments, and practising topics. Since study time is limited, it is important to decide which tasks should be prioritized and how time should be used effectively.

In this project, students will develop a C/C++ program called StudySmart AI. The system will apply algorithm design techniques, including Divide-and-Conquer, Greedy, and Dynamic Programming, to plan and optimize study tasks. A simple AI/ML model, such as Decision Tree or k-Nearest Neighbours, will also be used to recommend the most suitable algorithmic strategy based on task characteristics and time constraints.

This project allows students to apply algorithm analysis concepts to a realistic academic planning problem.

B. Course Learning Outcomes (CLO) covered

At the end of this assessment, students are able to:

CLO3 Demonstrate problem-solving skills via concepts, techniques, and proficiencies in algorithm analysis which leads to potential applications in the self-development stage e.g., further studies.

CLO4 Interpret the correctness of algorithms using mathematical proofs.

C. University Policy on Academic Misconduct

1. Academic misconduct is a serious offense in Xiamen University Malaysia. It can be defined as any of the following:
 - i. Plagiarism is submitting or presenting someone else's work, words, ideas, data or information as your own intentionally or unintentionally. This includes incorporating published and unpublished material, whether in manuscript, printed or electronic form into your work without acknowledging the source (the person and the work).
 - ii. Collusion is two or more people collaborating on a piece of work (in part or whole) which is intended to be wholly individual and passed it off as own individual work.
 - iii. Cheating is an act of dishonesty or fraud in order to gain an unfair advantage in an assessment. This includes using or attempting to use, or assisting another to use materials that are prohibited or inappropriate, commissioning work from a third party, falsifying data, or breaching any examination rules.
2. All assessments submitted must be the student's own work, without any materials generated by AI tools, including direct copying and pasting of text or paraphrasing. Any form of academic misconduct, including using prohibited materials or inappropriate assistance, is a serious offense and will result in a zero mark for the entire assessment or part of it. If there is more than one guilty party, such as in case of collusion, all parties involved will receive the same penalty.

D. Instruction to Students

1. Complete all tasks listed in the project requirements. All required program components, AI/ML module, study-task dataset, analysis, and documentation must be completed.
2. This assessment must be completed in groups of 4 to 5 students, subject to class size and lecturer approval.
3. Only the group representative should upload the submission files to Moodle.

Required Submission Files

a) Source Code

- Submit the .c or .cpp file(s).
- Ensure the code is clean, well-commented, organized, and executable.

b) Project Report (PDF format)

- Cover page with project title, group name, and student details.
- Contribution table outlining each group member's specific tasks.
- Description of the StudySmart AI scenario and dataset.
- Explanation of the implemented algorithms and AI/ML model.
- Results and comparison tables.
- Screenshots or evidence of program execution and testing.
- Individual component section for each student, clearly labelled with student name and ID.
- Well-commented source code in the appendix.
- Completed group and individual marking rubrics at the end of the report.

Note: Individual components should be included in the same group report. Separate individual files are not required unless instructed by the lecturer.

Submission Format and Deadline

- Upload all files as soft copy to Moodle as one compressed file. Only one submission per group is required.
- Students are required to submit the project report only through the assigned ASRP/Turnitin submission point for similarity checking. The source code is not required to be submitted for the similarity check.
- Deadline: All files must be submitted by: **Monday, June 29, 2026 (Week 13), 6:00 PM**. Late submissions will incur penalties.
- Use the following naming format:

YourGroupName_CST207_StudySmartAI_Project_202604.

D. Evaluation Breakdown

No.	Component Title	Marks	Percentage (%)
1.	Group Work Component	20	57
2.	Individual Work Component	15	43
	TOTAL	35	100

Table 1: Assessment Component Breakdown

F. Project Scenario

StudySmart AI simulates a smart academic planning application. The system receives a set of study tasks and helps the student decide which tasks should be prioritized and which planning strategy is most suitable. Each task has attributes such as study time, importance, deadline, difficulty, and task type.

The project simulates a real-world software development situation: imagine you are designing a smart study-planning application that automatically recommends the best algorithmic strategy based on task characteristics and student constraints.

G. Project Goal

The main goal of this project is to create a program that can:

- Generate or input different study-task scenarios.
- Implement multiple algorithmic strategies for task planning.
- Measure the performance and quality of each strategy.
- Extract features from the task scenario.
- Use an AI/ML model to predict the most suitable strategy.
- Compare the AI-predicted strategy with the actual best strategy.
- Present the results clearly using structured tables.

H. Required Study Task Data

Each group must use at least 8 to 15 study tasks. The tasks may be created manually or generated by the program. Each task should include the following fields:

Field	Description	Example
Task ID	Unique identifier for the study task	T1
Task Name	Name of the task or topic	Revise Dynamic Programming
Estimated Study Time	Time needed to complete the task	3 hours
Importance Score	Academic importance or expected benefit	5
Deadline / Urgency	How soon the task must be completed	2 days
Difficulty Level	Estimated difficulty of the task	1-5
Task Type	Lecture, tutorial, assignment, practice, or revision	Tutorial

Table 2: Study Task Data Fields and Descriptions

I. Group Component: System Development Task (57%)

Each group must design and implement the StudySmart AI system using C/C++. The system should contain the following required modules.

Module 1: Task Scenario Generation / Input

The system must allow Groups should prepare different scenarios, such as:

- Low-pressure scenario: available study time is enough for most tasks.
- High-pressure scenario: available study time is much lower than total required time.
- Deadline-focused scenario: several tasks have urgent deadlines.
- Importance-focused scenario: some tasks have much higher importance scores than others.

Each group must test **at least two different study-task scenarios** against all three algorithmic strategies (Modules 2, 3, and 4). The results of each scenario must be recorded in the performance comparison table (Module 6).

Module 2: Sorting / Divide-and-Conquer Module

Students must implement Merge Sort or Quick Sort (Choose one and justify your selection in the report) to rank study tasks based on one selected criterion, such as deadline, importance, difficulty, study time, urgency, or an importance-to-time ratio.

Rank	Task	Deadline	Importance	Study Time
1	Revise Dynamic Programming	2 days	5	3 hours
2	Complete Graph Tutorial	3 days	4	2 hours
3	Practise Greedy Problems	5 days	3	2 hours

Table 3: Example of Task Ranking Output (Sorting Module)

Module 3: Greedy Planning Module

Students must implement one greedy strategy to create a quick study plan. Possible greedy rules include highest importance first, earliest deadline first, shortest study time first, or highest importance-to-time ratio first. The report must explain the selected rule and discuss whether it always gives the optimal solution.

Module 4: Dynamic Programming Module

Students must implement the 0/1 Knapsack algorithm using Dynamic Programming. In this project, the available study time is the capacity. Each task has a required study time and an

importance score. The DP algorithm should select the combination of tasks that maximizes total importance without exceeding the available study time.

Selected Task	Study Time	Importance Score
Revise Dynamic Programming	3	5
Practise Graph Algorithms	4	6
Review Sorting Algorithms	2	3
TOTAL	9	14

Table 4: Example of Tasks Selected by Dynamic Programming Module

Optional Extension: Backtracking Approach: Backtracking is not required for the group implementation. However, students who choose Backtracking as their individual topic may explain and apply it to a small version of the study-task selection problem. For example, they may show how Backtracking explores possible task subsets and selects the best combination without exceeding the available study time. Since Backtracking has exponential time complexity, it should only be demonstrated using a small number of tasks.

Module 5: AI/ML Recommendation Module

Each group must include a simple AI/ML module that predicts the most suitable planning strategy for a given study-task scenario. The AI/ML module may be implemented using a simple Decision Tree, k-Nearest Neighbours (k-NN), or a manually designed rule-based decision tree. The model must be trained or built using a minimum of 8 to 15 labelled scenario examples, where each example consists of the extracted features listed in Table 5 and a known best strategy label. Students must include these training examples in the report to demonstrate how the model was built.

Note: The AI/ML component may be lightweight; students are not required to use advanced ML libraries, but they must clearly define the input features, prediction logic, and strategy classes.

The AI/ML module should use scenario features such as:

Feature	Description
Number of tasks	Total number of study tasks in the scenario
Total required study time	Sum of all task durations
Available study time	Total study time available to the student
Time pressure ratio	Total required time divided by available time
Average importance	Average importance score of all tasks
Deadline tightness	How close the deadlines are
Importance variation	Whether task importance scores are similar or very different

Table 5: Input Features for the AI/ML Recommendation Module

The model should predict one of the following strategy classes:

Prediction Class	Meaning
Sorting-based ranking	Suitable when the main goal is prioritization.
Greedy strategy	Suitable when a fast, simple plan is needed.
Dynamic Programming	Suitable when study time is limited and optimal task selection is important.

Table 6: AI/ML Strategy Prediction Classes

Module 6: Performance Measurement and Comparison

Students must compare the implemented strategies using suitable indicators. The comparison should include both practical results and algorithm analysis.

Students must complete this table with their results

Strategy	Selected Tasks	Total Study Time	Total Importance	Execution Time	Comment
Sorting-based ranking					
Greedy strategy					
Dynamic Programming					
AI/ML recommendation					

Table 7: Strategy Performance Comparison Summary.

Students must complete the strategy comparison table for each tested scenario. For each strategy, they should list the selected tasks, calculate the total study time used, calculate the total importance score, record the execution time, and add a short comment explaining the result. The

comparison should show which strategy performs better for the given scenario and why. Execution time can be measured using simple C/C++ timing functions, such as clock().

J. Individual Component (43% Marks)

Each student must write an individual section of approximately 3–4 pages, excluding tables and references. Each student must cite at least three credible references for the individual component, especially when discussing the modern or alternative algorithm. IEEE referencing style is recommended.

Each student must complete an individual section inside the group report. This section assesses the student's independent understanding of algorithm analysis and self-development through modern algorithm exploration. Each student must choose one algorithm used or related to the group project.

Each group member must select a unique algorithm or technique for the individual component that is used in or related to the group project, such as:

Student / Topic Option	Assigned Algorithm / Technique	Possible Modern / Alternative Algorithm
Option 1	Merge Sort / Quick Sort	TimSort, IntroSort
Option 2	Greedy task selection	Local Search, Approximation Algorithm
Option 3	0/1 Knapsack Dynamic Programming	Branch and Bound, FPTAS
Option 4	Backtracking	Branch and Bound, Constraint Programming
Option 5	Decision Tree / k-NN strategy prediction	Random Forest, Gradient Boosting

Table 8: Individual Component - Algorithm Topic Options

Required Individual Tasks

Each student must write the following five parts.

1. Explanation of Assigned Algorithm

Students must state the assigned algorithm and explain its purpose, main idea, input, output, and relevance to the StudySmart AI system in their own words.

2. Complexity Analysis

Students must provide the time and space complexity of the assigned algorithm, briefly explain how the complexity is derived, and relate it to the number of study tasks.

3. Application to StudySmart AI Scenario

Students must explain how the algorithm is used in StudySmart AI and support the explanation with a small example, table, trace, or step-by-step output.

4. Modern / Alternative Algorithm Research

Students must identify one relevant modern or alternative algorithm, explain its main concept, cite at least one credible reference, and discuss its possible benefits.

5. Comparison and Reflection

Students must compare the assigned algorithm with the alternative algorithm in terms of efficiency, scalability, difficulty, and suitability, then justify which is more appropriate for StudySmart AI.

k. Suggested Group Work Division

The project must be completed in groups of maximum 5 students. The following distribution is suggested, but groups may adjust it based on the number of members.

Student	Suggested Responsibility
Student 1	Study-task scenario design, dataset generation, and input handling
Student 2	Sorting / Divide-and-Conquer module
Student 3	Greedy planning module
Student 4	Dynamic Programming module
Student 5	AI/ML recommendation module, testing, result analysis, and report integration

Table 9: Suggested Group Work Division

Note: The AI/ML Recommendation Module is intended to be a lightweight strategy recommendation module, not an advanced machine learning task. Students are not required to use advanced AI/ML libraries. The module may be implemented using a simple manually designed rule-based decision tree in C/C++, where scenario features are used to recommend the most suitable algorithmic strategy.

MARKING RUBRICS

Component Title	Group Work					Percentage (20%)	Weight (%)	Marks
Criteria	Score and Descriptors							
	Excellent (9-10)	Good (7-8)	Average (4-6)	Need Improvement (3-4)	Poor (1-2)			
Study Task Dataset / Scenario Design (CLO3)	All required study-task data are complete, realistic, well-structured, and suitable for testing different algorithmic strategies.	Mostly complete dataset with minor missing details.	Basic dataset with limited attributes or weak scenario design.	Incomplete or unrealistic dataset.	Dataset/scenario is missing or not relevant to the project.	10		
Sorting / Divide-and-Conquer Module (CLO3)	Merge Sort or Quick Sort is implemented correctly and used effectively to rank tasks based on a clear criterion.	Sorting module works with minor logic, output, or efficiency issues.	Sorting module is partially correct but has noticeable implementation or ranking issues.	Sorting module is poorly implemented or only works for limited cases.	Sorting / Divide-and-Conquer module is not implemented.	10		
Greedy Planning Module (CLO4)	Greedy strategy is implemented correctly, clearly justified, and produces a reasonable study plan based on the selected rule.	Greedy strategy is mostly correct, with minor issues in justification, selection logic, or output.	Greedy strategy works partially but the selection rule or explanation is unclear.	Greedy strategy has major logic errors or is not properly connected to the study-planning problem.	Greedy module is not implemented.	10		
Dynamic Programming Module (CLO3)	0/1 Knapsack DP is implemented correctly to select the best combination of study tasks within limited study time.	DP module is mostly correct, with minor errors in table construction, output, or handling of edge cases.	DP module is partially correct but has some logic errors or incomplete explanation of selected tasks.	DP module has major errors and does not reliably solve the time-constrained task selection problem.	Dynamic Programming module is not implemented.	10		
	5	3-4	2-3	1	0			
AI/ML Recommendation Module (CLO3)	AI/ML model is clearly implemented or simulated using a valid simple model and predicts the most suitable planning strategy correctly in most test cases.	AI/ML module works with minor issues in feature selection, prediction logic, or explanation.	AI/ML module works partially but has limited examples, weak features, or unclear prediction process.	Poorly implemented, mostly rule-based without clear justification or predictions mostly incorrect.	AI/ML recommendation module is not implemented.	5		
	6-7	4-5	3-2	2-2	1-1	7		
Performance Measurement and Strategy Comparison (CLO4)	Results are clearly measured and compared using suitable indicators such as execution time, selected tasks, total importance, total study time, and prediction correctness.	Most results are measured and compared, with minor missing values or minor explanation issues.	Some comparison is provided but results are incomplete or not clearly analyzed.	Results are weak, inconsistent, or do not properly compare the strategies.	No meaningful performance measurement or comparison is provided.	7		
	5-5	4-4	3-3	2-2	1-1	5		

Code Clarity, Documentation, and Report Quality (CLO3)	Code is modular, readable, well-commented, and the report is well-organized with clear explanation, screenshots/tables, and complete sections.	Code and report are mostly clear, with minor formatting, documentation, or organization issues.	Understandable but incomplete documentation or explanation	Code/report are poorly organized, difficult to follow, or missing important sections.	Code/report are missing, unreadable, or not related to the project.	5	
					TOTAL	57	

Note to students: Please include this marking rubric at the end of the group report.

Component Title	Individual Work					Full Mark	Stud. ID	Stud. ID	Stud. ID	Stud. ID	Stud. ID
Criteria	Score and Descriptors										
	Excellent (5)	Good (4)	Average (3)	Need Improvement (2)	Poor (1)						
Explanation of Assigned Algorithm (CLO3)	Provides a clear, accurate, and complete explanation of the assigned algorithm, including purpose, main idea, input, output, and relevance.	Explanation is mostly clear and correct, with minor missing details.	Explanation is generally acceptable but lacks depth or has some unclear points.	Explanation is weak, incomplete, or shows limited understanding of the algorithm.	Explanation is missing or mostly incorrect.	9					
Complexity Analysis (CLO4)	Correctly analyzes time and space complexity with clear justification and appropriate discussion of cases where applicable.	Complexity analysis is mostly correct, with minor missing details or minor explanation issues.	Complexity is stated but explanation is limited or partially incorrect.	Complexity analysis is weak, unclear, or contains major errors.	Complexity analysis is missing or incorrect.	9					
Application to StudySmart AI Scenario (CLO3)	Clearly explains how the algorithm is applied in the StudySmart AI system, with a relevant example, table, or step-by-step illustration.	Application is mostly clear, with minor missing details or limited example.	Application is acceptable but general, with weak connection to the project scenario.	Application is vague or poorly connected to StudySmart AI.	No meaningful application to the project is provided.	9					
	7–8	5–6	3–4	1–2	0-1						
Modern / Alternative Algorithm Research (CLO3)	Identifies a highly relevant modern or improved alternative algorithm and explains its concept, purpose, and suitability clearly using appropriate references.	Alternative algorithm is relevant and mostly well explained, with minor missing details.	Alternative algorithm is identified but explanation is basic or only partially connected to the assigned algorithm.	Alternative algorithm is weakly related, poorly explained, or lacks credible support.	No suitable alternative algorithm is provided.	8					

Comparison and Reflection (CLO4)	Provides a clear and thoughtful comparison between the assigned algorithm and the modern alternative, discussing efficiency, scalability, implementation difficulty, and suitability.	Comparison is mostly clear, with minor missing aspects or limited reflection.	Comparison is acceptable but basic and lacks depth.	Comparison is weak, mostly descriptive, or does not clearly evaluate the two algorithms.	Comparison /reflection is missing or irrelevant.	8					
					TOTAL	43					

Note to students: Please include this marking rubric at the end of the individual report.